ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	The energy left is too little to support further levels.
2	**(d)** 5000 J	5 x 10 x 10 x 10.
3	**(b)** UV radiation on oxygen	O2 splits and recombines.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Decomposition recycles nutrients.
5	**(a)** 40 kJ	Two transfers.
6	**(b)** lower levels have more energy	Less loss in the chain.
7	**(d)** A is false but R is true.	A is false: energy flows in one direction; R is true.
8	**(b)** insects and eagles	Insects (T2) and eagle (T5).
9	**(a)** Both A and R are true and R is the correct explanation of A.	The simple substances released are taken up again by the producers.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R states the function of ozone, which is not the reason CFCs damage it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grass = 50 x 100 = 5000 J (1); Snake = 5 J (1).	1+1
12	Energy decreases to 10^{-9} of the original at the 10th level (1); Too little energy to sustain organisms (1).	1+1
13	Crop field has few species and needs inputs (1); Forest has many species and is self-sustaining (1).	1+1
14	Fox (T3) 80 J (1); Wolf (T4) = 8 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Small fish: 0.2 ppm (1); Big fish: 2 ppm (1); Kingfisher: 20 ppm (1).	1+1+1
16	Use both sides of paper (reduce) (1); Reuse bottles and bags (1); Recycle paper and glass; helps to conserve resources (1).	1+1+1
17	Toxic chemicals in effluents (1); Reduce oxygen levels (1); Poisoned food chain (1).	1+1+1
	OR: Food web has several connections (1); More predators or competitors for food (1); Chain reaction changes populations (1).	1+1+1
18	Releases toxic gases (1); Pollutes the air and harms organisms (1); Disturbs the food chains (1).	1+1+1
	OR: Vegetable waste decays (1); Plastic remains almost unchanged (1); Microbes decompose biodegradable matter only (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Phytoplankton (1). (ii) Third trophic level (secondary consumer) (1). (iii) 10% of 10,000 J is 1000 J (small fish), and 10% of this is 100 J (big fish) (1). (iv) 10 J (1); only about 10% of energy passes to the next level, the rest is lost as heat, so very little energy is left for further levels (1).	1+1+1+2
20	Energy available decreases by 90% at each step (1); Less energy can support less biomass (1); Top carnivores are few in number (1); They need a large area to find food (1); They are the most affected by loss of energy at lower levels (1).	1+1+1+1+1
	OR: Cereals: humans at the second level (1); Cattle: humans at the third level (1); If plants hold 10,000 J, humans eating cereals get 1000 J (1); Via cattle: cattle 1000 J, humans 100 J (1); So cereals feed about 10 times as many (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.
- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Grass, goat, human	Fewer levels lose less energy.
2	**(b)** protect the ozone layer	Chlorine is avoided.
3	**(a)** Both A and R are true and R is the correct explanation of A.	UV absorption is the reason.
4	**(b)** grass increases and carnivores decrease	Fewer plant-eaters and less prey.
5	**(d)** Composting	Gives manure.
6	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R is not an explanation of A.
7	**(b)** 1000 J	Three transfers.
8	**(a)** using paper bags and reusing bottles	Protects environment.
9	**(c)** A is true but R is false.	Only about 10 percent of the energy is passed on; the rest is lost, so long chains are not possible. R is false.
10	**(d)** A is false but R is true.	Ozone is formed when UV radiation splits O_2 into atoms; A is false, R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They convert dead organisms into simple inorganic substances (1); These are used again by plants (1).	1+1
12	Composting gives manure (1); Burning pollutes the air (1).	1+1
13	Pesticides get biomagnified (1); Top consumers hold the greatest concentration (1).	1+1
14	Producers: 10,000 J (1); Herbivores: 1000 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Forest: many species, complex food web (1); Crop field: few species, simple chain (1); Crop field is less stable and needs human inputs (1).	1+1+1
16	It is non-biodegradable (1); It persists in soil and water (1); It remains in food chains (1).	1+1+1
17	Herbivores would increase (1); Grass would be overgrazed (1); Eventually herbivores would starve (1).	1+1+1
	OR: They act on dead remains of all levels (1); They return nutrients to the soil (1); Producers need these nutrients (1).	1+1+1
18	It varies from organism to organism (1); Depends on the efficiency of use (1); But generally a small fraction is transferred (1).	1+1+1
	OR: Energy flows in one direction (1); It is lost as heat at each step (1); Materials, not energy, are recycled (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Biomagnification (1). (ii) Top consumers such as snakes and birds of prey (and humans, if they eat the grains and animals) (1). (iii) They are non-biodegradable and are not broken down or excreted, so they accumulate in the body (1). (iv) Organic farming, biological pest control, using less or safer pesticides, washing vegetables (any two) (1+1).	1+1+1+2
20	CFCs rise to the stratosphere (1); UV breaks CFCs and releases chlorine (1); Chlorine breaks down ozone to oxygen (1); Consequences: skin cancer, cataract, crop damage (1); Measures: UNEP agreement in 1987; individuals avoid CFC-based goods and aerosols (1).	1+1+1+1+1
	OR: (a) Toxic gases; remedy: recycle (1); (b) Ozone depletion; remedy: CFC-free products (1); (c) Water pollution, oxygen loss; remedy: sewage treatment (1); (d) Biomagnification; remedy: organic methods (1); Every action disturbs food chains (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.

- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.
- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.
- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** lose their food source	Dependent on the primary consumers.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Photosynthesis.
3	**(a)** biomagnification	Concentration increases up the chain.
4	**(b)** a fish-eating bird	Top.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Alternative paths.
6	**(b)** They can act at every trophic level on dead matter	They break down remains from every level.
7	**(b)** persist and accumulate	Remain in the environment.
8	**(a)** Both A and R are true and R is the correct explanation of A.	10 per cent law.
9	**(a)** Both A and R are true and R is the correct explanation of A.	This is biomagnification, and R explains it.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the second is a contrast and does not explain the first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They run off into water bodies (1); They enter food chains and biomagnify (1).	1+1
12	More UV reaches the surface (1); Harms phytoplankton, the producers, and the food chains (1).	1+1
13	Part is used in life processes like respiration (1); Part is lost as heat, so only 10% is passed on (1).	1+1
14	Many contain CFCs that harm ozone (1); They pollute the air (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(a) Many alternative pathways (1); (b) It biomagnifies and harms birds and humans (1); (c) Energy is not recycled and is lost as heat (1).	1+1+1
16	More demand for food (1); Plants and animals used excessively (1); Disturbs the balance and increases waste (1).	1+1+1
17	Producers are lost (1); The food chains that depend on them collapse (1); Soil is washed away and habitats are lost (1).	1+1+1
	OR: Very cold polar clouds help CFC reactions (1); Chlorine released destroys ozone (1); Sunlight in spring triggers the reaction (1).	1+1+1
18	Use cloth bags: no polythene waste (1); Use steel or glass bottles: reuse (1); Buy loose items: less packaging waste (1).	1+1+1
	OR: Organic waste is decomposed using oxygen (1); Oxygen level falls (1); Fish and other organisms die (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) O_3 (1). (ii) UV radiation splits O_2 into free atoms, which combine with O_2 to form ozone: $O_2 \longrightarrow O + O$; $O + O_2 \longrightarrow O_3$ (1). (iii) It absorbs the harmful ultraviolet radiation of the Sun (1). (iv) Skin cancer, cataract (damage to eyes), weakening of the immune system (any two) (1+1).	1+1+1+2
20	Initially herbivores increase (1); Plants are overgrazed (1); Food becomes scarce and herbivores starve (1); Disease and competition rise (1); So the claim is false; the balance is vital (1).	1+1+1+1+1

	OR: Producers: trees (1); Consumers: deer, rabbits, tigers, snakes; decomposers: fungi, bacteria (1); Chains: tree, deer, tiger and tree, rabbit, snake (1); Without decomposers dead matter and wastes would pile up (1); Nutrients would not be recycled and the producers would suffer (1).	1+1+1+1+1

Mistakes to avoid:

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